

NATURE INSPIRED

COMPUTING

TECHNIQUES

Assignment - 2

DNA Manipulation

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DNA Computing:

- (i) Known as molecular computing is a new approach to massively parallel computation based on groundbreaking work by Adleman.
- (ii) It was proposed as a means of solving a class of intractable computational problems in which the computing time can grow exponentially with problem size the 'NP Complete' or non-deterministic polynomial time complete problems).
- (iii) A DNA Computer is a collection of specially selected DNA strands whose combinations will result in the solution to some problem at hand.
- (iv) Currently available both to select the initial strands and to filter the final solution.
- (v) The main idea was the encoding of data in DNA strands and the use of tools from molecular biology to execute computational operations.
- (vi) For example, DNA computations may use a billion times less energy than an electronic computer while storing data in a trillion times less space.
- (vii) The two reasons for using molecular biology to solve problems is:
 - a) Information density of DNA is much greater than that of silicon.
 - b) Operations on DNA are massively parallel: a test tube of DNA can contain trillions of strands.

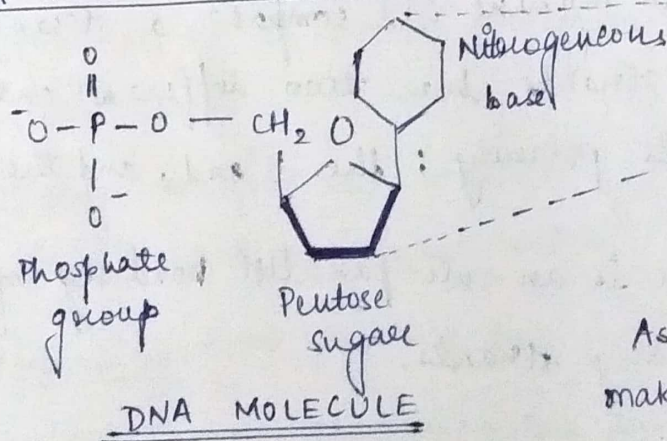
(ix) DNA supports two key functions for life:

1. coding for the production of proteins,
2. self replication

(x) Each deoxyribonucleotide consists of three components:

1. a sugar - deoxyribose
 - a. five carbon atoms: 1' to 5'
 - b. hydroxyl group (OH) attached to 3' carbon
2. a phosphate group.
3. a nitrogenous base.

DNA Molecule Fundamentals:



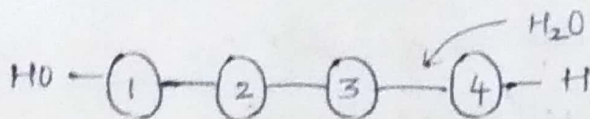
These components

- 1) Phosphate group
- 2) 'Pentose sugar'
- 3) Nitrogenous base

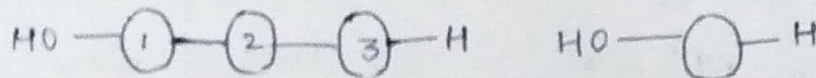
As a unit, these 3 components make up one monomer.

(i) A monomer is an individual unit that acts as a building block for large biological molecules. They are nearly identical units.

(ii) A polymer is a large molecule made up many small monomers. Think of a chain.



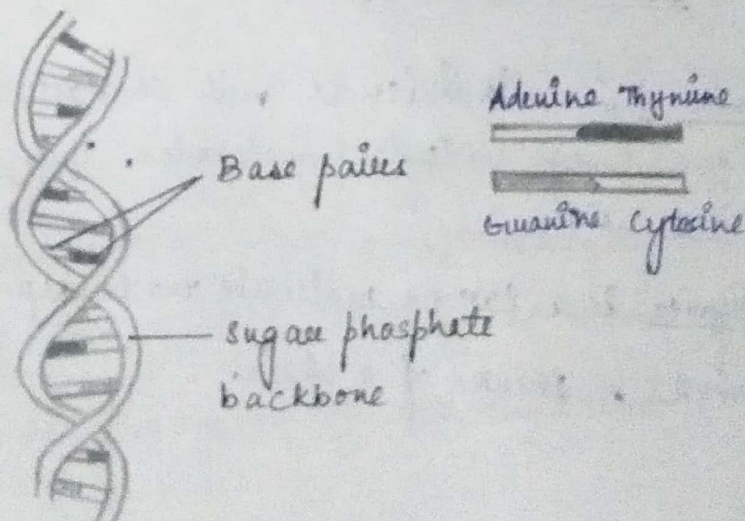
Hydrolysis adds a water molecule, breaking a bond



HYDROLYSIS OF A POLYMER

(ii) DNA monomers can link in two ways:

- The four nucleotides adenine (A), guanine (G), cytosine (C), and thymine (T) compose a strand of DNA. Each strand has two different ends that determine its polarity: the 3' end, and the 5' end.
- The double helix is an anti-parallel bonding of two complementary strands.



The Double Helix

PRINCIPLES OF DNA COMPUTING

- (i) A single strand of DNA is similar to a string consisting of a combination of four different symbols A, G, C, T.
- (ii) $\Sigma = \{A, G, C, T\}$ is needed to encode information which is more than enough considering that an electronic computer needs only two digits and for the same purpose.
- (iii) Computation takes place in test tubes or on a glass slide coated in 24K gold.
- (iv) Synthesizing a desired polynomial-length strand used in all models
- (v) Mixing: combine the contents of two test tubes into a third one to achieve union.
- (vi) Annealing: bond together two single stranded complementary DNA sequences by cooling the solution. Annealing in vitro is known as hybridisation.
- (vii) Melting: break apart a double-stranded DNA into its single-stranded complementary components by heating the solution. Melting in vitro is also known under the name of denaturation.

- (viii) Separating the strands by length using a technique called gel electrophoresis that makes possible the separation of strands by length.
- (ix) Extracting those strands that contain a given pattern as a selection by using affinity purification.
- (x) Cutting DNA double-strands at specific sites by using commercially available restriction enzymes.
- (xi) Ligating: paste DNA strands with compatible sticky ends by using DNA ligases.
- (xii) Substituting: inserting or deleting DNA sequences by using PCR site-specifications.

ADVANTAGES

- (i) Efficiently handle massive amounts of working memory.
- (ii) Expensive to build, being made of common biological materials.

DISADVANTAGES

- (i) Generating solution sets, even for some relatively simple problems, may require impractically large amounts of memory.
- (ii) DNA computers could not (at this point) replace traditional computers. They are not programmable and the average dumber can not sit down at a keyboard and get to work.